

Fixed Point Theory for the Bellman Operator

1 Motivation: why fixed points appear

In policy evaluation we derived the Bellman equation

$$v_\pi(s) = \sum_a \pi(a | s) \sum_{s'} p(s' | s, a) [\bar{R}(s, a, s') + \gamma v_\pi(s')].$$

Define the **Bellman operator** $\mathcal{T}_\pi : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$ by

$$(\mathcal{T}_\pi v)(s) = \sum_a \pi(a | s) \sum_{s'} p(s' | s, a) [\bar{R}(s, a, s') + \gamma v(s')].$$

Then the Bellman equation says exactly

$$v_\pi = \mathcal{T}_\pi v_\pi.$$

So v_π is a **fixed point** of $\mathcal{T}_\pi$. Fixed point theory answers whether such a point exists, whether it is unique, and how to compute it.

2 The setting: complete metric spaces

Definition 1 (Metric space). A metric space is a pair (X, d) where X is a set and $d : X \times X \rightarrow [0, \infty)$ satisfies

- $d(x, y) = 0 \iff x = y$,
- $d(x, y) = d(y, x)$,
- $d(x, z) \leq d(x, y) + d(y, z)$.

Definition 2 (Cauchy sequence). A sequence $\{x_n\}$ in (X, d) is Cauchy if for every $\varepsilon > 0$ there exists N such that $d(x_m, x_n) < \varepsilon$ for all $m, n \geq N$.

Definition 3 (Complete metric space). A metric space (X, d) is complete if every Cauchy sequence converges to a limit in X .

Remark. $\mathbb{R}^n$ with the sup norm $d(x, y) = \|x - y\|_\infty = \max_i |x_i - y_i|$ is complete. This is the space where value functions live.

3 The Banach fixed point theorem

Definition 4 (Contraction). Let (X, d) be a metric space. A map $T : X \rightarrow X$ is a **contraction** if there exists $\gamma \in [0, 1)$ such that

$$d(Tx, Ty) \leq \gamma d(x, y) \quad \text{for all } x, y \in X.$$

The constant γ is the **contraction factor**.

Theorem 5 (Banach, 1922). *Let (X, d) be a non-empty complete metric space and let $T : X \rightarrow X$ be a contraction with factor $\gamma < 1$. Then:*

1. **Existence and uniqueness.** T has exactly one fixed point $x^* \in X$: $Tx^* = x^*$.

2. **Convergence of iteration.** For any $x_0 \in X$, the sequence $x_{n+1} = Tx_n$ converges to x^* .

3. **Rate.** $d(x_n, x^*) \leq \gamma^n d(x_0, x^*)$.

4. **A priori bound.** $d(x_n, x^*) \leq \frac{\gamma^n}{1-\gamma} d(x_0, x_1)$.

5. **A posteriori bound.** $d(x_{n+1}, x^*) \leq \frac{\gamma}{1-\gamma} d(x_{n+1}, x_n)$.

Proof sketch. Existence. Pick $x_0 \in X$ and set $x_{n+1} = Tx_n$. For $m > n$,

$$d(x_n, x_m) \leq \sum_{j=n}^{m-1} d(x_j, x_{j+1}) \leq \sum_{j=n}^{m-1} \gamma^j d(x_0, x_1) \leq \frac{\gamma^n}{1-\gamma} d(x_0, x_1).$$

Since $\gamma < 1$, $\{x_n\}$ is Cauchy. By completeness, $x_n \rightarrow x^* \in X$. By continuity of T , $Tx^* = x^*$.

Uniqueness. If $Tx^* = x^*$ and $Ty^* = y^*$, then $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \gamma d(x^*, y^*)$, so $(1-\gamma)d(x^*, y^*) \leq 0$, hence $x^* = y^*$.

Rate. $d(x_n, x^*) = d(T^n x_0, T^n x^*) \leq \gamma^n d(x_0, x^*)$.

A posteriori. $d(x_{n+1}, x^*) \leq \gamma d(x_n, x^*)$ and $d(x_n, x^*) \leq d(x_n, x_{n+1}) + d(x_{n+1}, x^*)$, so $d(x_{n+1}, x^*) \leq \frac{\gamma}{1-\gamma} d(x_{n+1}, x_n)$. $\square$

4 Applying Banach to the Bellman operator

4.1 The space

Take $X = \mathbb{R}^{|S|}$ with the sup norm

$$\|v\|_\infty = \max_{s \in S} |v(s)|.$$

This is a complete metric space.

4.2 The operator is a contraction

Theorem 6. The Bellman operator $\mathcal{T}_\pi$ is a contraction on $(\mathbb{R}^{|S|}, \|\cdot\|_\infty)$ with factor γ :

$$\|\mathcal{T}_\pi u - \mathcal{T}_\pi v\|_\infty \leq \gamma \|u - v\|_\infty.$$

Proof. For any state s ,

$$\begin{aligned} |(\mathcal{T}_\pi u)(s) - (\mathcal{T}_\pi v)(s)| &= \left| \gamma \sum_a \pi(a | s) \sum_{s'} p(s' | s, a) (u(s') - v(s')) \right| \\ &\leq \gamma \sum_a \pi(a | s) \sum_{s'} p(s' | s, a) |u(s') - v(s')| \leq \gamma \|u - v\|_\infty \sum_a \pi(a | s) \sum_{s'} p(s' | s, a) = \gamma \|u - v\|_\infty. \end{aligned}$$

Taking the maximum over s gives the result. $\square$

4.3 Conclusion

By Banach's theorem, applied to $X = \mathbb{R}^{|S|}$ and $T = \mathcal{T}_\pi$:

1. There is a unique v_π with $v_\pi = \mathcal{T}_\pi v_\pi$.
2. For any v_0 , the iterates $v_{k+1} = \mathcal{T}_\pi v_k$ converge to v_π .
3. $\|v_k - v_\pi\|_\infty \leq \gamma^k \|v_0 - v_\pi\|_\infty$.
4. $\|v_{k+1} - v_\pi\|_\infty \leq \frac{\gamma}{1-\gamma} \|v_{k+1} - v_k\|_\infty$.

5 The optimality operator

Define

$$(\mathcal{T}_*v)(s) = \max_a \sum_{s'} p(s' | s, a) [\bar{R}(s, a, s') + \gamma v(s')].$$

Theorem 7. $\mathcal{T}_*$ is a contraction with factor γ in the sup norm.

Proof. Let $u, v \in \mathbb{R}^{|S|}$ and fix s . Let a^* achieve the max for u :

$$(\mathcal{T}_*u)(s) = \sum_{s'} p(s' | s, a^*) [\bar{R}(s, a^*, s') + \gamma u(s')].$$

Then

$$(\mathcal{T}_*v)(s) \geq \sum_{s'} p(s' | s, a^*) [\bar{R}(s, a^*, s') + \gamma v(s')],$$

so

$$(\mathcal{T}_*u)(s) - (\mathcal{T}_*v)(s) \leq \gamma \sum_{s'} p(s' | s, a^*) (u(s') - v(s')) \leq \gamma \|u - v\|_\infty.$$

By symmetry, the same bound holds with u and v exchanged. Hence

$$|(\mathcal{T}_*u)(s) - (\mathcal{T}_*v)(s)| \leq \gamma \|u - v\|_\infty.$$

Taking the maximum over s gives the result. □

So $\mathcal{T}_*$ has a unique fixed point v_* , and value iteration $v_{k+1} = \mathcal{T}_*v_k$ converges to it geometrically.

6 Consequences and interpretation

6.1 Existence and uniqueness

Banach guarantees that v_π and v_* exist and are unique without solving any linear system.

6.2 Convergence of iterative methods

Iterative policy evaluation, value iteration, and their asynchronous variants converge because they are (exact or approximate) applications of a contraction.

6.3 The role of $\gamma < 1$

If $\gamma = 1$, the operator need not be a contraction and the theory breaks down. For $\gamma = 1$, one needs ergodicity or stochastic shortest path theory.

6.4 The role of completeness

Completeness of $\mathbb{R}^{|S|}$ under $\|\cdot\|_\infty$ makes the Cauchy sequence converge. For infinite state spaces, use $\mathcal{B}(\mathcal{S})$, the Banach space of bounded functions with the sup norm.

7 Extensions

7.1 Infinite state spaces

Take $X = \mathcal{B}(\mathcal{S})$ with $\|v\|_\infty = \sup_s |v(s)|$. Banach applies verbatim.

7.2 Weighted sup norms

Use $\|v\|_w = \sup_s |v(s)|/w(s)$ for a weight $w > 0$. Under suitable conditions, the Bellman operator is a contraction in this norm too.

7.3 Stochastic operators

For TD learning, the operator is random. One uses stochastic approximation theory: the expected operator is a contraction, and the noise is handled by Robbins–Monro conditions.

7.4 Other fixed point theorems

- **Brouwer:** continuous maps on compact convex sets have a fixed point (no uniqueness, no iteration).
- **Kakutani:** set-valued generalization, used in game theory.
- **Schauder:** infinite-dimensional version of Brouwer.
- **Knaster–Tarski:** monotone maps on complete lattices.

Banach is preferred for MDPs because it gives uniqueness and a constructive iteration with a rate.